

FermBench: A new benchmark for measuring the capabilities of LLMs on fermentation knowledge

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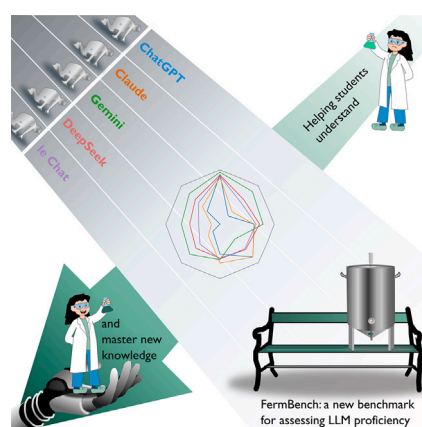
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HIGHLIGHTS

- This work introduces FermBench, a new dataset on fermentation containing both human-curated and AI-generated responses.
- FermBench is used to benchmark LLMs powering highly used chatbots, including ChatGPT, Gemini, DeepSeek, Claude and le Chat.
- The experimental framework includes various evaluation methods: automated metrics, human annotations and LLM-as-a-Judge.
- Results suggest that the knowledge currently embedded in these models is satisfactory; no LLM was significantly better.
- Students should follow institutional/governmental guidelines and personal preferences informed by the identified parameters.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Artificial intelligence
Large language models
Fermentation
Chatbots in education

ABSTRACT

Generative Artificial Intelligence (GenAI) chatbots continue to amaze users worldwide with their rapid improvements. These tools possess vast general knowledge and can thus be used in various fields, including education. However, before rolling out these models in pedagogical applications, it is fundamental to understand whether the information provided is reliable and if any of the currently available chatbots are best suited for domain-specific tasks. The objective of this study is to thoroughly investigate these aspects in a specific domain, fermentation, with the overarching goal of providing guidelines to students and teachers to select the best GenAI assistant. To achieve this goal, we introduce *FermBench*, a dataset specifically designed for fermentation processes. We use the collected data to benchmark five large language models (LLMs) powering commercially available GenAI chatbots, including ChatGPT, Gemini, DeepSeek, Claude and le Chat. To evaluate the responses of these models, we propose a robust experimental framework that includes automated metrics, human annotations, and the LLM-as-a-Judge approach. The obtained results suggest that, given the high baseline and the fact that the judges were unable to agree on an overall best model, the current knowledge embedded within these models is adequate and the

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standalone results cannot provide pedagogical guidelines regarding which chatbot should be used in education. These results suggest that the choice of which GenAI chatbot should be supported by institutional or government guidance, as well as individual preferences, perhaps informed by the parameters identified in our analysis. An interesting finding of the study is that curated answers are not necessarily better than generated ones.

1. Introduction

Recent advances in artificial intelligence (AI) present educators with the potential to digitize course materials, thus enriching both students' educational experiences and their engagement (Kasneji et al., 2023). This transformation has been enabled by the rise of commercially available generative AI (GenAI) chatbots, such as ChatGPT and Microsoft Copilot. Despite initial skepticism, these tools have been increasingly used in different fields and applications, including higher education. Large language models (LLMs) are the foundational models that power most GenAI chatbots; they are deep learning models that are designed to understand and generate human language (text). Recent studies demonstrated the effectiveness of LLMs as educational tools, indicating a likely future role for LLMs in engineering education (Kamalov et al., 2023). LLMs are capable of delivering customized student support (Rodrigues et al., 2024), enhancing understanding and academic success. Research further indicates that AI can substantially boost learning effectiveness by managing cognitive load autonomously, offering individualized instruction, and flexibly adjusting learning trajectories (Gkintoni et al., 2025). Research highlights that AI-powered tools are especially beneficial in domains with high cognitive demands, such as STEM, where they can greatly improve understanding, retention, and problem-solving capabilities (Gkintoni et al., 2025).

However, with many models being available and the constant release of new updates that continuously challenge the state-of-the-art (SOTA) and beat the predecessors and the competitors, a few questions arise:

- Are these models good enough for higher education?
- And if they are, is the difference in performance significant or can students and educators freely choose the model of their preference?

This study aims to address these questions and try to fill the gap in the current knowledge that different LLMs possess, ultimately evaluating their suitability for educational contexts. More specifically, we assess which commercially available chatbot provides better responses on a specific topic and, therefore, could better support student learning. The selected topic is *fermentation* and the target student groups are chemical engineering students. To this end, we selected the most widely known and used GenAI chatbots currently available. The selected chatbots are: ChatGPT, Claude, Gemini, DeepSeek and le Chat. To evaluate the generated responses, we introduce *FermBench*, a new benchmark dataset of open-ended fermentation questions, which we release publicly. The performance of the selected models is compared through a series of quantitative and qualitative analyses, including validation by field experts. The experts are asked to rank the completions, which are the output responses generated by the LLMs, as well as to answer the questions themselves. This procedure enables us to compare the generated answers with the curated ones. Moreover, we also compare the evaluations provided by the field experts to the ones generated by LLMs used as judges (LLM-as-judge approach), to verify if humans and LLMs agree on the overall quality of the responses. Finally, curated answers are used to enrich the knowledge of a previously implemented custom LLM, FermentAI (Caccavale et al., 2024), embedded in an educational e-learning platform, BioVL (Caño de las Heras et al., 2022), which is publicly available and can be used to ask questions about fermentation processes.

The remainder of the manuscript is structured as follows. In Section 2, a short review of the literature related to the selected chatbots and their extent of usage in education is presented. Subsequently, the

data collection, dataset curation process and the experiment design are introduced in Section 3. In Section 4, the results of the experiment are presented, with quantitative and qualitative comparisons of the models. Moreover, we also discuss the curated answers provided by expert annotators, and compare them to LLM generations. Finally, we discuss the key insights of the experiment in Section 5, highlighting the strengths and weaknesses of each GenAI model, as well as possible future perspectives (Section 8) and limitations (Section 6).

2. Background and aim

This section aims to provide a contextualization of the work and the current state-of-the-art (SOTA). It first provides a brief overview of the development of the GenAI chatbots selected to be benchmarked in this work, including a comparison of their similarities and differences. Then it presents other studies that have benchmarked various LLMs for similar applications and domains. Afterward, we discuss some examples of these models that were applied to educational contexts, and to (bio)chemical engineering education in particular, since this is the field of the proposed benchmark. Finally, we introduce the chatbot that is the focus of improvement in this work.

2.1. Brief history of the selected GenAI models

This section provides an overview of the release dates of the selected commercial chatbots. They are presented in chronological order.

ChatGPT was created by OpenAI and released in November 2022. The chatbot is based on the Generative Pre-Trained Transformers (GPT) series of foundational LLM models (Brown et al., 2020; Radford et al., 2018). In Q1 2025, the free version of the chatbot was based on GPT-3.5 turbo. Users had access to GPT-4o (Hurst et al., 2024) with usage limits, and when those limits were reached, the model was automatically switched to a less capable model (GPT-3.5 turbo). In August 2025, OpenAI began the rollout of the GPT-5 models to power the chatbot, making GPT-5 the new default model for all logged-in users. For free users, once the usage limits are reached, the application switches to using GPT-5 mini. Undeniably, ChatGPT was the model that started the "AI revolution", leading to unprecedented accessibility and adoption of this technology and to a massive shift in public engagement with AI (Mesko, 2023).

Claude was created by Anthropic and released in March 2023 (Anthropic, 2023a). The chatbot interfaces with three LLMs: Haiku, Sonnet, and Opus. The latest release, as of Q1 2025, was Claude Sonnet 3.7 (Anthropic, 2023b). In May 2025, Anthropic released Sonnet 4.

Gemini was created by Google AI and released in December 2023 (although it was first launched in March 2023 as Bard) (Google, 2023). It is based on the Gemini foundational LLM series created by Google DeepMind (including Gemini 1.0, Gemini Ultra, Gemini Pro, Gemini Nano and Gemini 2.0) (Google, 2023). In Q1 2025, Gemini's free version of the AI assistant was powered by Gemini 2.0 Flash. In June 2025, the 2.5 Flash version was released for stable use and integrated into the chatbot.

DeepSeek chatbot was created by DeepSeek and was released in January 2025 (DeepSeek, 2025), although the foundational models were released in November 2023 and the first chatbot was released in November 2024 (DeepSeek, 2024). In Q1 2025, the DeepSeek chatbot was powered by DeepSeek-V3. An update for increased reasoning capabilities, DeepSeek-R1-0528, was released in May 2025. The most recent major release, DeepSeek-V3.1, was launched in August 2025. This

Table 1

Selected GenAI models specifications. As of April 2025, the online chatbot interfaces of the selected models are free; however, some of the APIs used for the completion process require paid credits. OS stands for open-source.

Chatbot	Developer	Country	Model	OS	Model name in API	API Tier
ChatGPT	OpenAI	USA	GPT-3.5	No	gpt-3.5-turbo	Paid
Claude	Anthropic	USA	Claude 3.7 Sonnet	No	claude-3-7-sonnet-20250219	Paid
Gemini	Google	USA	Gemini 2.0 Flash	No	gemini-2.0-flash	Free
DeepSeek	DeepSeek	China	DeepSeek-V3	Yes	deepseek-chat	Paid
le Chat	Mistral AI	France	Mistral Large	No	mistral-large-latest	Freemium

version was post-trained on the original V3 base, with a focus on hybrid inference, faster thinking, and stronger agent skills.

Le Chat was created by Mistral AI and released in February 2025, although the beta version was available since February 2024 (MistralAI, 2025a). The model used in the free version of the chatbot until Q1 2025 was Mistral Large. Since then, new models with enhanced reasoning capabilities have been released, such as the Magistral Small and Medium and Voxtral Small and Mini. Mistral Medium 3.1 was released in August 2025. It is described as a frontier-class multimodal model, improving tone and performance (MistralAI, 2025b).

Microsoft Copilot is another widely used chatbot. Microsoft Copilot is used by many institutions (such as the University responsible for this study) as a reliable model, with stipulated contracts on data sharing through institutional access. However, it was not selected for this study due to the fact that it has no publicly available application programming interface (API). It was created by Microsoft and released in February 2023. It was originally called Bing Chat. It was then rebranded in September 2023 as Microsoft Copilot (Microsoft, 2023). Microsoft Copilot is built upon OpenAI's GPT-4 series of LLMs; however, the exact backend model is not disclosed. Worth noting is that, although OpenAI remains an entirely independent company governed by the OpenAI Nonprofit (OpenAI, 2025), Microsoft has a multibillion-dollar investment in OpenAI.

An overview of the models is provided in Table 1, including the specific APIs used in this work. All of the selected models are Transformer (Vaswani et al., 2017) based generative LLMs, so they necessarily share some similarities, but also present some differences. All models have a decoder-only architecture, although Gemini incorporates significant modifications for multi-modality. All models selected are text-only except Gemini 2.0 Flash, which is multi-modal, allowing images, audios, and videos to be uploaded and created.¹ Gemini and Mistral (le Chat) use sparse Mixture-of-Experts (MoE), meaning that the model has multiple "expert" subnetworks, and for a given input, only a subset of these experts are activated. This significantly improves the capacity, performance and efficiency of the model. GPT-3.5 is also believed to use this architecture, although this is not officially confirmed by OpenAI.

In order to make these models more ethical, OpenAI started applying a reward mechanism described as Reinforcement Learning from Human Feedback (RLHF). To extend on this, Anthropic introduced Constitutional AI, which is a process involving a supervised learning (SL) and a reinforcement learning (RL) phase. In the SL phase, an initial model is sampled from, then self-critiques and revisions are generated, and then the original model is fine-tuned on revised responses. In the RL phase, the fine-tuned model is sampled from, and then a model is used to evaluate which of the two samples is better. This process leads to the training of a preference model from the dataset of AI preferences. The model is then trained with RL using the preference model as the reward

signal. The result is an AI assistant that engages with harmful queries by explaining its objections (Bai et al., 2022).

2.2. GenAI chatbots in education

With the growing number of AI assistants available, it is not surprising that students' interest in using these tools has increased, especially to facilitate learning. As a consequence, recent research has begun to methodologically investigate students' use of GenAI chatbots. Although the results vary between studies and disciplines, the use of chatbots appears to be widespread among students (Stöhr et al., 2024). For example, Deschenes and McMahon (2024) report that nearly 65% of undergraduate and graduate students at Harvard University have used or plan to use GenAI chatbots for academic work. Regarding the effect of AI chatbots on learning and performance, Stöhr et al. (2024) show that 47.7% of the respondents agreed that the chatbots they used made them more effective learners, while only 17.3% confirmed the positive effect of chatbots on improving their grades. The study also finds that over 50% of students are positive about chatbots but have concerns about their potential use. Similar results are also found in Caccavale et al. (2024). Furthermore, Deschenes and McMahon (2024) show that most of the respondents (65%) do not find the responses of GenAI chatbots trustworthy enough for academic purposes, adding that students actively use these tools, but want guidance to use them more effectively. It is therefore evident that, although GenAI chatbots can positively influence the learning process and aid students with fast responses, more research is required to ensure the correctness, trustworthiness and reliability of the generated responses and to further investigate which chatbot is better in specific domains.

2.3. GenAI chatbots in (bio)chemical engineering education

Similarly to other domains, researchers have recently started using GenAI tools in (bio)chemical engineering education. Tsai et al. (2023) investigate the integration of ChatGPT into chemical engineering education to enhance problem-solving and programming skills, proposing an LLM-assisted problem-solving procedure that moves beyond traditional classroom-based learning by providing hands-on experience and fostering critical thinking. Keith et al. (2025) designed a laboratory session where first-year students utilized ChatGPT to develop a model predicting water drainage from a tank, which they then experimentally validated. An example of a custom chatbot that was implemented for a specific application is represented by Caccavale et al. (2025), where an LLM performing retrieval-augmented generation (RAG) has been developed for an M.Sc. course at the Technical University of Denmark.

Narrowing the scope further, the use of GenAI chatbots appears to still be scarce in fermentation. Fermentation holds significant importance in (bio)chemical engineering education because of its diverse applications in the production of renewable fuels, biochemicals, biopharmaceuticals, food and beverages. Understanding the fundamentals of fermentation enables students to develop innovative and sustainable methods that tackle environmental issues while enhancing economic efficiency within the (bio)chemical industry. In educational curricula, students learn about catalysis and the biochemical pathways necessary

¹ The paid version of ChatGPT is built upon GPT-4-turbo, which does allow for multi-modal input/output, but it was not selected for this study as the interface is not free.

for implementing efficient and effective production systems related to fermentation-derived compounds (Doménech et al., 2022). Therefore, by gaining expertise in fermentation processes, students acquire the essential knowledge and skills to shape the future of green chemistry and bioengineering, thereby aiding in the development of more sustainable industrial practices. An example of GenAI tools to aid learning in fermentation is Ferment AI (Caccavale et al., 2024).

2.4. Available benchmarks

Robust evaluation is an essential aspect of developing safe and ethical LLMs. However, in order to validate these models, large-scale, high-quality data are a requirement.

Various benchmarks are available that can be used to evaluate LLMs in different domains. For example, MMLU (Hendrycks et al., 2020) covers 57 tasks that span various domains, from simple mathematics to complex legal reasoning, and enables the measurement of the multitask accuracy of LLMs to help researchers pinpoint important shortcomings of these models. GLUE (Wang et al., 2018) and SuperGLUE (Wang et al., 2019) are collections of tools for evaluating natural language understanding capabilities of models, including a diverse set of tasks and a variety of metrics (e.g., accuracy, F1 score, and correlation coefficients) tailored to each task to ensure a comprehensive assessment of model performance. TruthfulQA (Lin et al., 2021) is a benchmark devised to measure whether an LLM is truthful in question-answering (QA); the model is evaluated through a generation task, which aims to maximize truthfulness and ensure informativeness, and a multiple-choice task, to identify true statements among multiple options.

Although the aforementioned benchmarks are generally useful in assessing various capabilities of LLMs, such as QA reliability, factual accuracy and informativeness, they do not necessarily provide a reference point for students on a specific topic. Different studies have compared the knowledge of some commercially available chatbots on domain-specific topics. For example, Laurent et al. (2024) proposes a benchmark for biology research and measures the performance of several models against it, reporting results compared to human expert biology researchers. Another model comparison, where GPT-4o is compared to Claude 3.5 Sonnet, is provided by Mitchener et al. (2025), where the authors also introduce BixBench, a benchmark dataset for bioinformatics. However, more research should be done to help students (and educators) understand whether different chatbots provide significantly different responses or if the tool selection should simply follow personal preference or institutional guidelines.

2.5. Custom LLM

Another question that should be answered is whether generalized LLMs are “good enough” to answer specialized questions, for example, in the area of fermentation, or whether custom chatbots should be implemented. Among the available custom chatbots, there is FermentAI (Caccavale et al., 2024), a chatbot previously developed by the authors. It is based on a RAG approach (Lewis et al., 2020), which is the process of optimizing the output of an LLM by enriching the given prompt. This process enables the model to reference a knowledge base outside of its training before generating a response. This allows the LLM to extend its knowledge without being re-trained, therefore ensuring a cost-effective approach for LLM output improvement. FermentAI leverages the Flan-T5 (Chung et al., 2024) pre-trained sequence-to-sequence model. The prompt of the LLM is composed of a question and enriched by its context, the curated information for the model to generate the answer.

3. Methods

Many available benchmark datasets are based on binary or multiple-choice answers (e.g., MMLU). Others focus on general intelligence (e.g., TruthfulQA and MMLU); therefore, generalizing on a domain-specific

application could be challenging. Even domain-specific benchmarks (e.g., BixBench) might not generalize to a similar domain, although within STEM. Additionally, evaluating answers to open-ended questions is often a challenge. Since there is no universal and accepted ground truth, as is often the case with binary or multiple-choice questions, it is often difficult to find a reliable source of information against which to compare AI-generated answers in domain-specific applications. To address this challenge for a specific domain, fermentation, we propose a dataset, *FermBench*, that contains curated answers to open-ended questions. In this study, we benchmark the five LLMs selected; the responses generated by the LLMs are then evaluated by experts. The experts are recruited from the authors’ professional network and hold (or are pursuing) a PhD in the field of chemical engineering, having worked extensively with fermentation. The experimental framework is shown in Fig. 1. The figure presents the process adopted in this study, from data collection (questions asked by a group of experts), curation of the question list by the authors, and the answer generation of the various LLMs. The completions were validated using three methods: (i) automatic curation, (ii) human annotations (by a different group of experts), and (iii) LLM-as-a-Judge approach.

It should be noted that throughout the article, we often refer to the LLMs used for the experiments with the name of the corresponding chatbot available (e.g., GPT-3.5 vs ChatGPT). We do this consciously because we want students and educators to identify with the results and the findings, rather than only appealing to the technical AI community.

3.1. Data acquisition

We asked participants of two training schools, “Mechanistic Modelling in Bioprocess Engineering: From Theory to Application” (31st March – 3rd April 2025) and “Advanced Use of BioVL and E-Learning Tools” (4th April 2025) under the DIG4BIO initiative, to fill in a survey meant to collect questions regarding fermentation.

The participants were informed about the experiment as follows: “*This form is created to collect questions about fermentation. We are working on developing digital tools to teach fermentation, and so we need to collect data and curate it for this purpose. This is why we would like to ask you to write 10 or more questions about fermentation that you think would be valuable for a digital tool (such as a chatbot, for example) to answer. These questions will then be answered by experts and used to develop these digital tools. The tool will be used in an educational context, to help students learn (or deepen their knowledge) about fermentation. The questionnaire is anonymous and we do not collect emails, therefore the evaluation will not, in any way, matter your final grade. By participating, you agree that these results will be used for scientific research.*”.

Additionally, the task to be performed is described below: “*Please suggest questions about fermentation that you think students should have knowledge about and that are important for them to be able to answer. When thinking about these questions, ask yourself: (i) “What is important to know about fermentation?”, (ii) “What should a student learn?”, (iii) “What should a student be able to answer?”. You can ask basic, intermediate and more complex questions.*”.

This resulted in a dataset of 118 questions, which were carefully reviewed for relevance and/or repetitions.

3.2. GenAI model selection

Five commercially available GenAI models were selected for this experiment. The motivation behind the selection varies among the models and will be addressed in this section.

ChatGPT was selected because it gained worldwide attention and was arguably the first model to raise awareness about GenAI models among the wider population, sparking reflections on their potential use in education. Claude and Gemini were selected because they generally achieve good results on used benchmarks (Anthropic, 2024; Ramachandran, 2024). DeepSeek and le Chat were selected due to the fact that they are

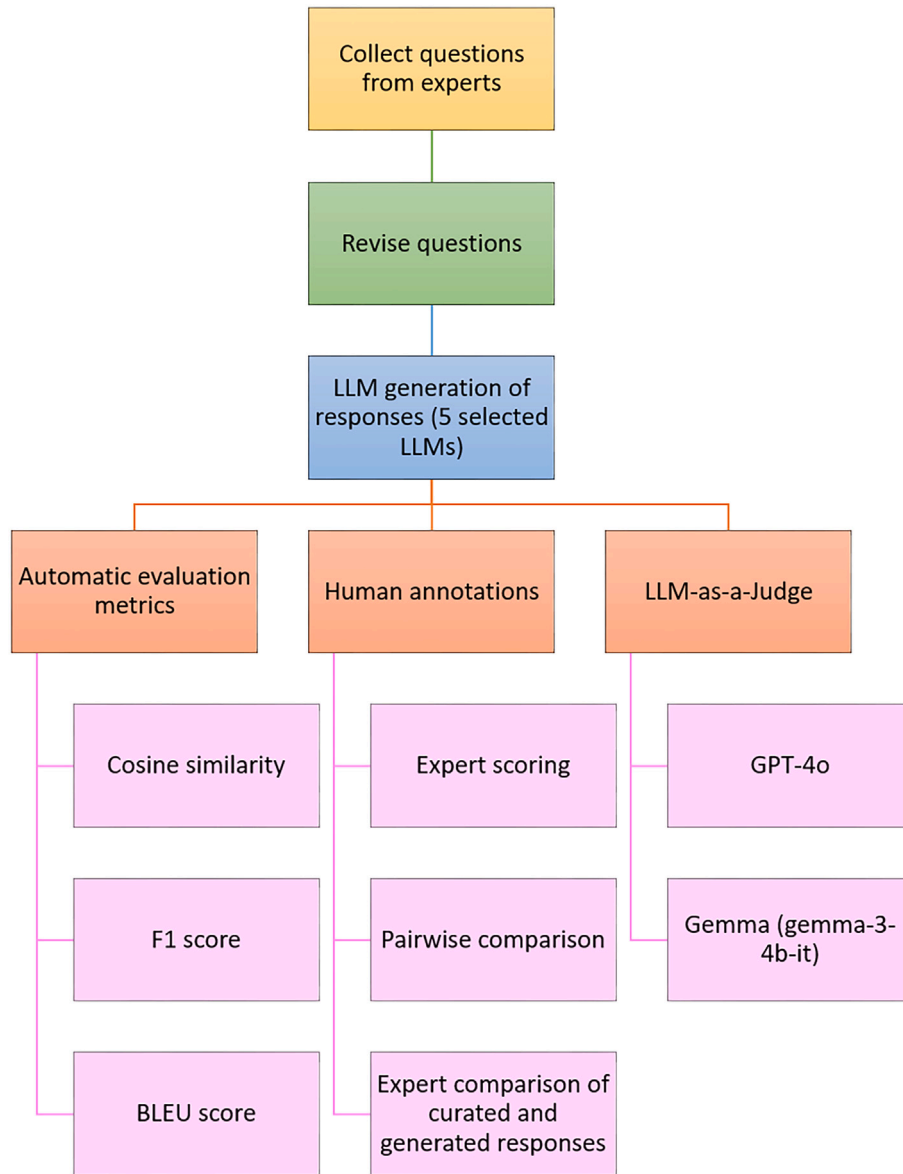


Fig. 1. Schematics of the experimental framework used in this study, starting with the data collection (questions asked by a group of experts), the curation of the question list by the authors, and the answer generation of the various LLMs. The completions were validated using three approaches, (i) automatic evaluation metrics, (ii) human annotations, and (iii) LLM-as-a-Judge approach.

developed in countries other than the United States of America (USA), specifically in China and France.

We employed the models used in the non-paid version of the various chatbots at the time of the experiment (April 2025), e.g., although GPT-4o was available, we used GPT-3.5-turbo for this work, since it was the model leveraged by the free version of ChatGPT.² The reasoning behind this choice is that the overall intent of the paper is to provide a reference for students who are trying to learn about fermentation, and therefore might not be willing or able to subscribe to the upgraded versions (i.e., the latest releases). However, the models used in the free ChatGPT

(GPT-3.5-turbo), Claude (claude-3-7-sonnet-20250219) and DeepSeek (deepseek-chat) chatbots required paid tokens to access their APIs; therefore, we paid for the API calls. We did not select the deep thinking and reasoning versions of the models.

3.3. Completion process

We interfaced with various LLMs by using their APIs, which act as standardized digital messengers to send requests and receive structured data from the models. Further specifications on the models are provided in Table 1. We generated the completion of all the questions collected in *FermBench* for every model. Every set of responses (one per model) was saved as a JSON Lines file, where the key is the question and the value is the generated answer. For ease of use, the files were then joined and saved as a spreadsheet.

Table 2 presents further information regarding the length of the generated responses, indicated by the average number of tokens, for

² As of April 2025, the free tier of ChatGPT users have limited access to GPT-4o. Once the limit is reached, the chatbot likely switches to GPT-3.5, which is the next available model for free-tier users. While GPT 3.5 is still a very capable model, it may not perform as well as GPT-4o, particularly in more complex or nuanced tasks.

Table 2

Descriptors by token. The average length for both ChatGPT and Claude is approx. 200, while in Gemini it is more than quadruple. DeepSeek and le Chat, on average, produce, respectively, 394 and 405 tokens per sentence. Here, *tokens* is intended as a word, not how each LLM tokenizes sentences internally.

Chatbot	Avg. tokens
ChatGPT	205.66
Claude	198.80
Gemini	830.63
DeepSeek	394.26
le Chat	404.55

each LLM. As shown in the table, there is a substantial difference in completion length among the selected models.

3.4. Automatic evaluation metrics

The performance of the models is assessed through a series of metrics. We consider both N-gram based metrics, such as Bilingual Evaluation Understudy (BLEU) score (Papineni et al., 2002) and the F1 score, and semantic-based metrics, such as cosine similarity. Cosine measures, such as similarity and distance, are commonly used in natural language processing (NLP) to assess the relationship between words or sentences. BLEU is a precision-based metric originally developed for evaluating machine translation to quantify the alignment between the generated and the reference text (Papineni et al., 2002). It does this by counting matching N-grams (sequences of n words). The score ranges between 0 and 1, where a higher score represents a greater degree of similarity. The F1 score is the harmonic mean of precision and recall score and compares the set of predicted tokens. Automated evaluation metrics can inadvertently introduce biases that do not accurately reflect the real-world performance of a model. For example, N-gram based metrics can favor certain linguistic structures or overlook nuances in language understanding (Dubois et al., 2024). Additionally, these metrics have been criticized for not always correlating well with human judgments, especially with respect to semantic similarity and fluency, and can be sensitive to word order and might not capture the nuances of meaning (Liu, Iter, et al., 2023; Reiter, 2018; Tan et al., 2015; Tran et al., 2019).

In order to provide a more visual understanding of how the completions are related, we propose three methods for dimensionality reduction: principal component analysis (PCA) (Pearson, 1901), t-distributed stochastic neighbor embedding (T-SNE) (Van der Maaten & Hinton, 2008) and uniform manifold approximation and projection (UMAP) (McInnes et al., 2018). The results are averaged across three model runs and different random seeds to improve the validity and robustness of results (Biderman et al., 2024; Sellam et al., 2021).

3.5. Expert annotation and evaluation process

In this study, the standard metrics used regularly to assess LLMs' generations—cosine similarity, F1 score and BLEU score—are only meant to provide an exploratory analysis, but are not sufficient alone to establish which model might be more accurate. In order to achieve that, it is necessary to establish a clear benchmark with ground truth observations, against which the output of the models can be compared. Therefore, in addition to these metrics, we ask human experts ($N=9$) to provide responses to the questions collected in *FermBench* and to evaluate the answers given by the selected LLMs. This is done in a three-step process, plus an additional pairwise comparison, which will be explained in Section 3.6. For each question in a set of given questions ($N=15$), we ask experts to:

1. Answer the questions with no supporting aid (this is not used to evaluate their responses, but serves as a baseline for the curated answer collected afterwards).
2. Score the responses given by the LLMs according to the following parameters: factual accuracy, comprehensiveness, coherence to the question asked, concision, readability, helpfulness, insightfulness and overall quality. The experts are asked to score each response according to these parameters on a Likert scale (1-5, where 5 is best). Given that the workload for the human annotators would have been too significant if asked to annotate the complete dataset, we randomly selected a total of 15 questions. To verify the representativeness of these questions, we performed a topic analysis: we confirmed that the 15 questions follow the same topic distribution as the overall dataset.
3. Provide a revised response to the given question, where they can use all aids, including the responses given by the LLMs.

The given parameters are selected to assess various aspects of the responses. Factual accuracy aims to assess to what extent the answer is correct or if the model is hallucinating. Comprehensiveness measures the completeness of the response, whether it answers the question fully or if there are missing elements. Coherence focuses on whether the response tackles the question and aligns with user intent. Concision evaluates whether the response is redundant or contains superfluous information. Readability provides an overview of how easy the response is to read and understand, or if it is too complex, and if the language is fluent and grammatically correct. Helpfulness evaluates whether the answer is useful to the user/student. Insightfulness assesses the depth of the answer and whether it provides novel perspectives or goes beyond a superficial response. Overall quality provides a general assessment of the response.

To calculate an overall score, we add the scores for each parameter to obtain the points for every response. We then calculate the average for each model.

3.6. Pairwise comparison

Finally, to further validate the scoring obtained from the previous setup, we ask experts to perform a small-scale ($N=10$) pairwise comparison of responses. These questions were the same for all annotators. For 5 LLMs ($n=5$), the number of unique pairs is given by the formula $n(n-1)/2 = 10$ unique pairs.

3.7. Annotator agreement

To quantify the agreement between annotators, and therefore to conclude whether the results can be considered statistically significant or should be taken more as a general trend, we calculate the inter-annotator agreement. The inter-annotator agreement is measured as Krippendorff's α (Krippendorff, 2011).

3.8. LLM-as-a-judge

Generally, human/expert evaluation is the gold standard for assessing human preferences. However, this approach has some limitations, including the fact that it is extremely slow and costly (Zheng et al., 2023). To address this challenge, a method that is starting to be commonly used in order to evaluate open-ended, generated responses from chatbots is to use SOTA LLMs as a surrogate for humans. Because these models are often trained with RLHF, they already exhibit strong human alignment. An example of this approach is provided by Zheng et al. (2023), where the authors note that existing benchmarks often fail to capture human preferences for open-ended tasks. This could be mitigated by using strong LLMs, such as GPT-4, as automated judges to assess other LLMs. The study proposes two new benchmarks and concludes that LLM-as-a-judge is a scalable and explainable method for approximating human preferences, which can otherwise be costly to obtain. However, this approach also presents some biases and limitations, including position, verbosity, self-enhancement biases, and limited reasoning ability

(Zheng et al., 2023). According to Wang et al. (2023), ChatGPT shows promise as a general-purpose natural language understanding evaluation metric, but its reliability may vary depending on the methods of constructing the dataset. Thakur et al. (2024) analyze the alignment and vulnerabilities of various LLMs-as-judges, highlighting biases and limitations compared to human evaluators. Gu et al. (2024) suggest that LLMs can be used as judges for complex tasks, but ensuring reliability requires careful design and standardization.

To provide further evaluation of the responses generated by the five selected LLMs, as well as to further compare human evaluations, we also adopt the LLM-as-a-Judge approach. We select two LLMs, Gemma (gemma-3-4b-it) (Keith et al., 2025) and GPT-4o (Hurst et al., 2024), and use the same task description for the evaluation as used with the human experts. The models used as judges have substantially different sizes: GPT-4o being likely the largest model used in this study, and Gemma being the smallest. This was chosen to compare the results and assess whether the model size would make a difference in this task.

3.9. Statistical analysis

To assess the performance of the five LLMs on the FermBench dataset ($N = 118$), we applied a two-stage, non-parametric statistical procedure (Demšar, 2006). In the first stage, we ran a Friedman test (Friedman, 1937, 1940) for each metric to detect overall differences in model performance while accounting for the blocked design (all models evaluated on the same 118 questions). In contrast to ANOVA, the Friedman test operates on ranks, which makes it well-suited for handling the ordinal and potentially non-normal distribution of LLM-as-a-judge scores. For metrics where the Friedman test revealed significant differences ($p < 0.05$), we then carried out a Nemenyi post-hoc test (Nemenyi, 1963). This follow-up test pinpoints which model pairs differ significantly while controlling the family-wise error rate in the presence of multiple comparisons. All analyses were performed separately for each of the two judges (Gemma and GPT-4o) to examine inter-rater agreement and the sensitivity of each judge.

3.10. Evaluating curated responses and improvement of custom LLM

To evaluate the quality of the responses provided by the human annotators (experts), we perform an additional experiment where we compare them to randomly selected generated responses. We recruit one annotator who was not involved in the previous annotation process, and therefore is not aware of which responses were generated and which were given by experts. We then calculate the frequency (in %) with which the annotator preferred curated responses over the completions of the chatbots.

The responses collected in this study, both generated and curated, are then reviewed by the authors. The resulting data, composed of question-answer pairs, is used to enrich the dataset used by our custom LLM presented, *FermentAI*.

4. Results

This section presents the results of the study. We will introduce (i) metric comparison of the completions, including cosine similarity, F1 and BLEU score; (ii) experts' evaluations, both for scoring and pairwise experiments; and (iii) the results for the LLM-as-a-Judge benchmark.

4.1. Automatic metric comparison of completions

The comparison of the generated responses is presented in Table 3 and visualized in Fig. 2, and includes cosine similarity, F1 score, and BLEU score. The results are averaged over three runs. Table 3 and Fig. 3 intend to provide a preliminary analysis of possible shared syntax and semantic connotations among models. This comparison is also important to understand whether models are consistent in their responses and therefore the general meaning of the answer stays the same, or whether there is within-model stochasticity. The latter would mean that the responses of the models could vary too much to be reliable for further comparisons with other models.

The results suggest that, reasonably, the responses generated by each model are more similar to each other within the same model than to

Table 3

Reported metrics for the selected GenAI chatbots. The first value is the pairwise cosine similarity (c sim), the second is the F1 score, and the third value is the BLEU score. All metrics are averaged over three runs.

Chatbot	ChatGPT			Claude			Gemini			DeepSeek			le Chat		
	c sim	F1	BLEU	c sim	F1	BLEU	c sim	F1	BLEU	c sim	F1	BLEU	c sim	F1	BLEU
ChatGPT	0.90	0.29	0.33	0.81	0.17	0.10	0.79	0.12	0.08	0.81	0.14	0.07	0.83	0.18	0.15
Claude	0.81	0.17	0.10	0.89	0.32	0.31	0.80	0.12	0.06	0.83	0.17	0.10	0.82	0.16	0.10
Gemini	0.79	0.12	0.08	0.80	0.12	0.06	0.86	0.21	0.28	0.82	0.14	0.09	0.81	0.15	0.13
DeepSeek	0.81	0.14	0.07	0.83	0.17	0.10	0.82	0.14	0.09	0.91	0.30	0.34	0.84	0.17	0.14
le Chat	0.83	0.18	0.15	0.82	0.16	0.10	0.81	0.15	0.13	0.84	0.17	0.14	0.88	0.25	0.30

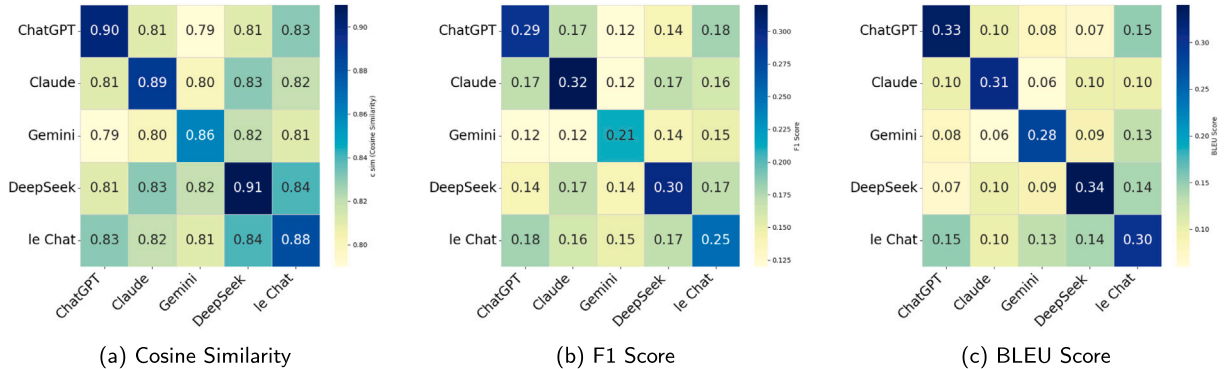


Fig. 2. Heatmaps for, respectively, cosine similarity, F1 score and BLEU score of the selected GenAI chatbots.

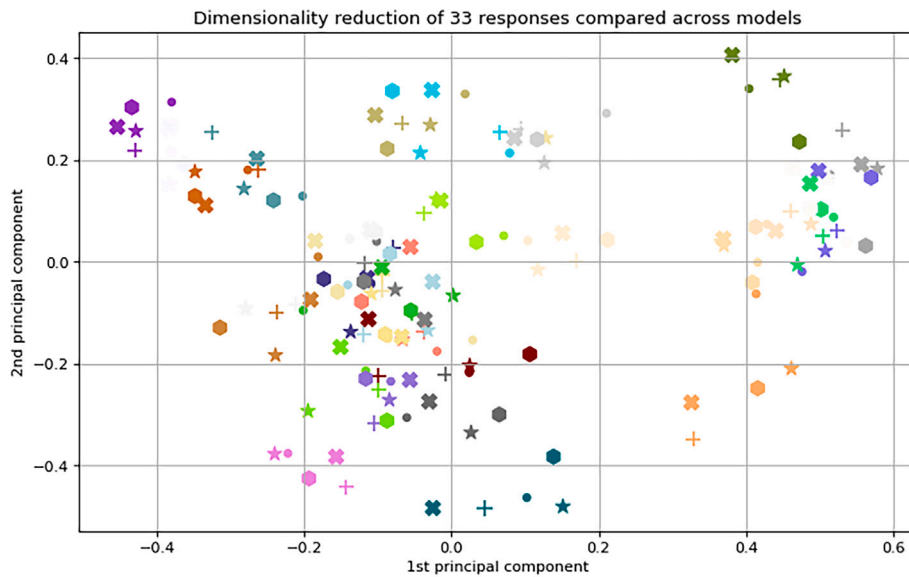


Fig. 3. Visualization of the first two principal components of the responses of the selected models. ChatGPT is represented by the “X” symbol, Claude by the hexagon, Gemini by the dot, DeepSeek by the “+” sign and le Chat by the star. The colors represent different responses. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4

Evaluation of *FermBench*. Comparison of the responses of each chatbot by field experts. The last column represents the average of all the parameters. The results are based on 15 randomly selected questions. For each question, the 9 annotators recruited in this study were asked to assess the responses of each of the 5 LLMs selected ($N=75$). Asterisks denote statistical significance relative to ChatGPT based on Nemenyi post-hoc tests (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

Chatbot	Fact. Acc.	Compr.	Cohér.	Concis.	Readab.	Helpful.	Insight.	Ov. Quality	Avg.
ChatGPT	4.72	3.40	4.18	4.61	4.04	3.35	3.00	3.63	3.87
Claude	4.61	3.60	4.19	4.61	4.35**	3.57	3.30	3.86	4.01
Gemini	4.80***	4.59	4.49	3.73	4.23	4.48**	4.34*	4.50***	4.40
DeepSeek	4.70***	4.18	4.37	4.25	4.18	4.18	4.05	4.21***	4.27
le Chat	4.66**	4.04	4.25	4.21	4.21	3.98*	3.74***	4.05	4.14

other models, such that two different completions of DeepSeek to the same prompt are more similar to each other (cosine similarity of 0.91, F1 score of 0.30 and BLEU of 0.34) than to a completion by le Chat (cosine similarity of 0.84, F1 score of 0.17 and BLEU of 0.14). The generations from DeepSeek seem semantically more similar than other models (e.g., the meaning of the sentences is closer) since the cosine similarity is the highest (0.91); however, the syntactic similarity (e.g., the consistency in the language and vocabulary used, denoted by the F1 score since it is used as a token-level similarity measure) is higher in Claude (0.32), while other models tend to paraphrase and vary their vocabulary more with each model run. In terms of lexical and structural similarity, which the BLEU score can be an indication of, DeepSeek presents the highest (0.34). Responses by ChatGPT are more similar, on average, to responses by le Chat than by other models, such as Claude, Gemini, and DeepSeek. Fig. 3 shows the semantic relationship (closeness of the responses) across the five groups in a shared space. Here, we create a single array that contains all the sentence embeddings from all five response datasets. Then we applied the dimensionality reduction technique (in this case, PCA (Pearson, 1901)) to this combined dataset. This approach focuses on finding the principal components that best capture the overall variance and structure across the datasets, therefore resulting in a single transformed space where all sentences are embedded. Visualizations of other dimensionality reduction methods, T-SNE and UMAP, can be found in the Appendix, Figs. 7 and 8.

4.2. Evaluation by field experts

Table 4 shows the evaluation of the field experts for the responses generated by the different models. Overall, experts seem to prefer the responses provided by Gemini (avg. 4.4), with DeepSeek (avg. 4.27) being the second-best. Information provided by ChatGPT (avg. 3.87) gained the lowest average score. Gemini scores highest in most of the parameters: factual accuracy (4.80), comprehensiveness (4.59), coherence (4.49), helpfulness (4.48), insightfulness (4.34) and overall quality (4.50). Claude is found to be the most readable (4.35) and, along with ChatGPT, the most concise (4.61 for both). However, this conciseness leads to a compromise in all other parameters, since these two chatbots score lower than all the others.

Statistical significance was calculated by the Friedman test, followed by the Nemenyi post-hoc test. These tests ensure that two tests aim to calculate with mathematical certainty that two models are different. For both human evaluators and LLM-as-a-Judge scores, the statistical significance is relative to ChatGPT (i.e., is there a statistical difference between Gemini and ChatGPT in factual accuracy). There are cases where certain models seem to obtain substantially higher performance (e.g., DeepSeek in insightfulness), however this may not be statistically significant due to high variance in the answers (for example, if DeepSeek gave 5/5 answers to 30 questions but 2/5 answers to the other 26, its average stays high, but the test cannot prove it is significantly better than ChatGPT).

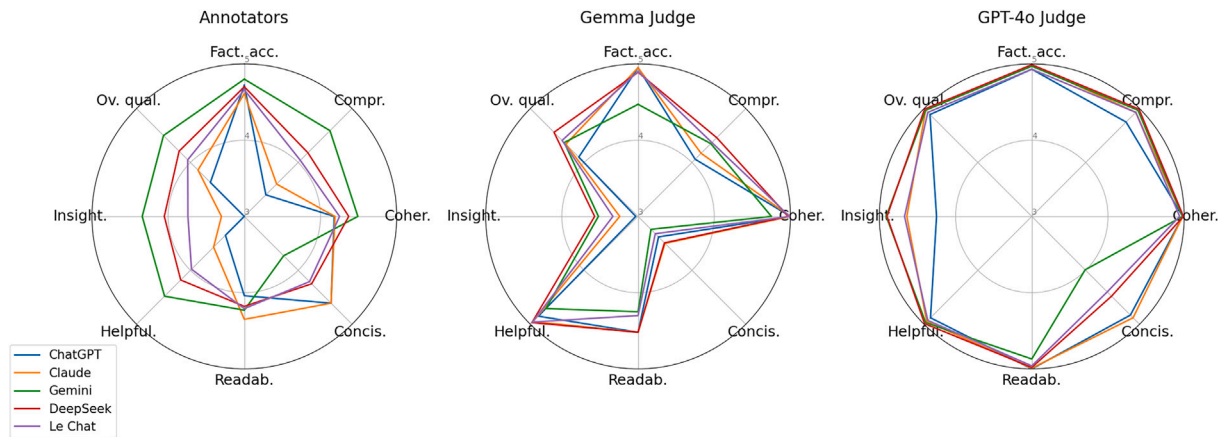


Fig. 4. Radar charts of the results, including expert scoring (first graph on the left) and LLM-as-judges (Gemma and GPT-4o). The axis ranges from 3 to 5 to highlight the difference. The center point is the lowest score (in this case approx. 3). The outer edge of the chart is the highest possible score (5 in the Likert scale). For example, the first chart shows a great difference between Gemini (green) and ChatGPT (blue) regarding insightfulness, where the first is evaluated to be more insightful than the latter. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.3. LLM-as-a-judge scores

Another strategy used to compare the completion of the various models was to use the LLM-as-a-Judge approach. Following the same script given to the experts, two LLMs, Gemma (gemma-3-4b-it) and GPT-4o, were asked to score the responses according to the given parameters. Gemma scores DeepSeek highest across almost all parameters (avg. 4.43) except for factual accuracy (4.89), coherence (4.97) and readability (4.52, on par with ChatGPT and Claude), where in all three cases, Claude and ChatGPT obtain the same score. Overall, after DeepSeek, the highest-scoring model is Claude (avg. 4.33), then le Chat (avg. 4.32), ChatGPT (avg. 4.24) and finally Gemini (avg. 4.21). This result is also surprising given that Gemma is created by Google DeepMind and built from the research and technology used to create Gemini models (Keith et al., 2025). Similarly, GPT-4o scores the responses generated by ChatGPT (GPT-3.5) the lowest (avg. 4.81). This LLM-as-a-Judge scores completions by Claude the highest (avg. 4.92), followed by DeepSeek (avg. 4.91), le Chat (avg. 4.84) and Gemini (avg. 4.83). However, if we disregard the total average and consider the overall quality parameter, the two LLMs agree that DeepSeek produces the best responses (avg. 4.99), followed by Claude (avg. 4.97) and Gemini (avg. 4.97) with the same score, then le Chat (avg. 4.93) and finally ChatGPT (avg. 4.89).

Some of the results are statistically significant. Both Gemma ($p < 0.001$) and GPT-4o ($p < 0.01$) rated Gemini and DeepSeek as significantly more insightful than the ChatGPT baseline. This agreement across judges points to a stable *technical tier*, in which these models deliver more advanced scientific reasoning. At the same time, the outcomes reveal judge-specific sensitivities. Gemma was more attuned to overall quality and comprehensiveness, selecting Gemini and DeepSeek as the top systems. Because Gemma's ratings were not tightly clustered near 5.0, it retained enough statistical resolution to detect these differences. GPT-4o, in contrast, was particularly sensitive to concision ($p \approx 10^{-10}$). It sharply penalized Gemini (3.99) and DeepSeek (4.48) relative to ChatGPT (4.83), indicating a strong preference for the more compact style characteristic of the OpenAI and Anthropic models. Neither judge detected significant differences in factual accuracy, coherence, or readability. This implies that, within the FermBench domain, all evaluated models are operating at a *performance ceiling* for basic linguistic form and factual recall, and that meaningful distinctions now emerge primarily in "higher-order" dimensions such as insightfulness and comprehensiveness. An important observation was the *ceiling effect*

in the GPT-4o evaluations: although GPT-4o found global significance for metrics like overall quality ($p < 0.001$), the ratings were so tightly compressed (all around ~ 4.9) that pairwise Nemenyi tests could not identify a clear single winner. This underscores the value of employing smaller, more 'critical' models like Gemma as judges to preserve evaluative granularity.

4.4. Comparing expert and LLM scores

Although the judges (both human and LLMs) present some differences in scoring the selected chatbots across the various parameters used, overall, our study reveals strong agreement between human expert annotators and automated LLM-based evaluators ($p < 0.05$ for the main metrics). The Gemma judge's *high-resolution* assessment aligned more closely with human experts' fine-grained judgments than the more *leniency-biased* GPT-4o. Moreover, the consistent ranking of Gemini and DeepSeek as the top models for technical fermentation insights across all three evaluator groups offers robust support for the validity of the FermBench framework. It should also be noted that LLMs tend to judge concision more harshly than annotators, leading to longer answers, such as those generated by Gemini, le Chat, and DeepSeek, which receive lower scores and, consequently, impact the overall result.

Fig. 4 shows the difference between the average parameters annotated by experts (first graph on the left), compared to the two LLMs-as-judges, Gemma (second graph) and GPT-4o (third graph). The plot highlights that, while GPT-4o tends to evaluate various LLMs higher and more similarly, the annotators are able to detect more nuances in the responses. Notably, the annotators find a bigger gap between the models regarding comprehensiveness, helpfulness, and insightfulness. Gemma also presents less variation than the annotators, but is more strict with the scores for conciseness and insightfulness.

4.5. Results of the pairwise comparison

For the expert annotations, the trend presented in Table 4 is also confirmed in the pairwise comparison, shown in Table 6. In this experiment, experts were asked to select the best answer out of two given options. Experts confirmed their preference for responses generated by Gemini, the highest-ranking chatbot, followed by DeepSeek, le Chat, Claude and ChatGPT. On the other hand, the LLM-as-a-Judge chosen for this task, GPT-4o, is not consistent with the previous experiment. While in Table 5 GPT-4o scored higher responses from Claude, the highest

Table 5

Evaluation of *FermBench*. Comparison of the responses of each chatbot by other LLMs used as judges. The last column represents the average of all the parameters. The results are averaged over the whole dataset (N=118). Asterisks denote statistical significance relative to ChatGPT based on Nemenyi post-hoc tests (*p < 0.05, **p < 0.01, ***p < 0.001).

Judge	Chatbot	Fact. Acc.	Compr.	Coher.	Concis.	Readab.	Helpful.	Insight.	Ov. Quality	Avg.
Gemma	ChatGPT	4.95	4.06	5.00	3.38	4.52	4.85	3.03	4.10	4.24
	Claude	4.95	4.17	5.00	3.50	4.52	4.95	3.24	4.34	4.33
	Gemini	4.47	4.35**	4.75	3.24	4.25	4.71	3.52***	4.37***	4.21
	DeepSeek	4.89	4.46*	4.97	3.49	4.52	4.97	3.57***	4.56***	4.43
	le Chat	4.90	4.37	4.99	3.32	4.30	4.96	3.33	4.41	4.32
GPT-4o	ChatGPT	4.93	4.75	4.98	4.83	5.00	4.88	4.25	4.89	4.81
	Claude	4.99	4.93	4.98	4.88	4.99	4.95	4.64	4.97	4.92
	Gemini	4.97	4.97	4.98	3.99***	4.87	4.97	4.91**	4.97	4.83
	DeepSeek	4.99	4.99	4.99	4.48*	4.98	4.99	4.90*	4.99	4.91
	le Chat	4.93	4.93	4.94	4.40	4.96	4.93	4.67	4.93	4.84

Table 6

Pairwise comparison of chatbots' responses by field experts and GPT-4o as LLM-as-a-Judge. For the human annotations, the results are calculated from the answers of 9 expert annotators over 10 questions. The win percentage is normalized according to the total number of responses collected (N=86). For the LLM, the results are averaged over 4 runs.

Chatbot	Wins	Percentage [%]	GPT-4o [%]
ChatGPT	10	12	6.7
Claude	14	16	3.3
Gemini	27	31	20
DeepSeek	19	22	40
le Chat	16	19	30

Table 7

Pairwise comparison of generated and curated responses. The results are calculated over 15 questions. The win percentage is normalized according to the total number of responses collected (N=15). The percentage of the last row is adjusted to ensure the sum of all rounded values equals the target total (100).

Chatbot	Wins	Percentage [%]	Total [%]
ChatGPT	3	100	
Claude	0	0	
Gemini	2	66.7	66.7
DeepSeek	3	100	
le Chat	2	66.7	
Annotators	4	26.7	26.7
None	1	6.6	6.6

values are obtained for, respectively, DeepSeek, le Chat and Gemini, while responses from Claude are selected the least amount of times.

However, these results should be interpreted as preliminary observations, since the annotator agreement for the pairwise experiment is 0.083, while for a subset of scoring questions (7x160) it is 0.043. The α coefficients suggest that, although agreement is slightly better than chance (higher than 0), the annotators do not have substantial agreement.

4.6. Are curated responses actually better?

In the expert annotation and evaluation process, described in Section 3.5, in addition to scoring the generated responses, experts were also asked to answer the questions themselves. We then selected the best answers, which we will refer to as "curated answers" and, in this task, asked another expert annotator who was not involved in the previous

annotation process to select the best response out of two given options. In this case, for each question (N=15), one response is generated by an LLM (three randomly selected responses for each model used in this study) and the other is curated by an expert. Table 7 shows that, as indicated by the high scores in the previous tasks, chatbots generally generate satisfactory responses, even outperforming annotators.

Table 8 contains the expert's justifications on why some answers, both generated and curated, are preferable. Among the reasons behind selecting the human annotations, for example, is the fact that some generated responses tend to be too descriptive or conceptually incorrect (e.g., first answer). However, this descriptiveness can be positive, since human annotations sometimes tend to write too short responses, leaving out important information. Chatbot responses are also found to be more structured than responses provided by the annotators.

5. Discussion

The impact of introducing AI in education is a noteworthy topic to be reflected upon before deploying these models and using them with students. It has been thoroughly investigated in many studies, including Caccavale et al. (2024, 2025); Yan et al. (2024). Therefore, it will not be further elaborated here, although we briefly discuss it in Section 6.

This section aims to reflect on the experiences presented in this work. To this end, it is divided into three parts, where the following is discussed: (i) working with human annotators; (ii) trends in the generations and main takeaways; and (iii) contributions and overall reflections.

5.1. Working with human annotators

One caveat of LLMs is that the data used to train these models is not publicly available. This means that guaranteeing the correctness and fairness of the source material used can be challenging. When working in a field such as education, ensuring that responses are correct is of the utmost importance. So, it would be preferable to have a mechanism to address this challenge, for example, through a human-in-the-loop approach or by having a RAG system with access to data manually curated by experts.

However, relying on human annotations can be costly and laborious (Zheng et al., 2023). In this work, recruiting human annotators, creating a form for them to rate responses, and getting them to respond to the survey took a significant amount of time. We estimated the work to be approximately 3 hours per annotator, and we were able to recruit 9 annotators. That meant a total of 27 hours of work. We gave annotators a time frame of 7–10 days to complete the surveys. However, most of the annotators took longer to respond, delaying the data collection process. This process was originally estimated to last 15 days. The total time spent collecting responses amounts to approximately 70 days. For reference, the process of writing the code and computing the responses

Table 8
Justifications provided by the expert for a response selection.

Winner	Response IDs	Expert justification
B	A [Gemini] B [Annotator]	"A is too descriptive, and in fact contains some statements that are not correct. In A, it is stated several times that models will optimize processes. That is conceptually wrong. Models can be used to optimize a process, when combined with expert knowledge or with an optimization algorithm."
B	A [Annotator] B [DeepSeek]	"A is very short, and A does not really mention whether it is about developing a first principles, a hybrid or a data-driven model."
B	A [ChatGPT] B [Annotator]	"B is short and to the point!"
B	A [Annotator] B [Claude]	"B is more concise, more to the point"
B	A [le Chat] B [Annotator]	"Difficult choice. Again, B is more to the point according to me."
B	A [Annotator] B [Gemini]	"B is much longer, but it gives a very detailed explanation, and [...] more clarity compared to A"
A	A [DeepSeek] B [Annotator]	"A is more detailed. I like the introductory statement of B more, but otherwise I think that information in A is better structured!"
B	A [Annotator] B [ChatGPT]	"I like the more descriptive approach taken in B!"
A	A [Claude] B [Annotator]	"A is more detailed. But there is a fundamental error in the response because you confuse reactor type (stirred tank, bubble column, packed bed) with reactor operating modes in a single list. This numbered list should be separated into two separate numbered lists!"
A	A [Annotator] B [le Chat]	"B is more extended, but I think that the simple explanation given in A could be sufficient!"
A	A [Gemini] B [Annotator]	"A is far more detailed than B, and that is probably needed for a field that is as complex as fermentation. [...]"
B	A [Annotator] B [DeepSeek]	"A is too simple, B is more detailed. The first sentence in A also sounds like something is missing!"
A	A [ChatGPT] B [Annotator]	"A is more detailed and more to the point."
B	A [Annotator] B [Claude]	"What I don't like about A is that it does not state the possibility that one can also expand on the Monod model, by expanding it towards a more complicated kinetic behaviour, to describe more complex biological phenomena."
None	A [le Chat] B [Annotator]	"A and B are equally bad in stating that you only produce CO ₂ and H ₂ O. That is wrong. Many products of interest are produced using aerobic conditions. For example enzymes at [...] are typically produced under aerobic conditions!"

for the five selected LLMs (with responses to the given questions re-run three times to account for uncertainty) lasted approximately 10 times less. Furthermore, not all experts completed the annotation process, resulting in fewer responses than originally estimated, as further discussed below.

A visual summary of the annotation process, highlighting the "annotation dropout" is presented in Fig. 5. During the initial phase of the experiment (E1), we randomly selected a set of 37 questions, which we categorized into three types: question type A (QTA) for scoring parameters, question type B (QTB) for open-ended questions, and question type C (QTC) for pairwise analysis. However, we realized early in the experiment that this task was too long for the annotators (EA),

prompting us to reduce it to 15 questions for the subsequent iteration (E2). Unfortunately, even with the reduced setup, not all annotators completed all questions. Consequently, the final number of responses was QTA = 310, QTB = 115, and QTC = 86, respectively. This process turned out to be longer (and perhaps more frustrating) than originally anticipated, making it challenging to complete the study. Additionally, this is a very fast growing field, meaning that, besides the fact that we have fewer responses than we expected to have, the delay also impacted the study significantly.

Of note is that human annotators were recruited from the professional network of the authors, which means that other researchers attempting the same task may have more (or fewer) problems in finding experts willing to participate in the experiment. This variability of human resources must also be openly discussed in this line of research.

Disclosing these aspects aims to highlight the trade-off between, on one hand, the desire to conduct research with the best intentions and address the known challenges with data quality and response correctness of commercially available LLMs, and, on the other hand, the practical limitations that working with human annotators implies. This trade-off must be addressed by the community, and better solutions need to be found, such as establishing a network of expert annotators available to researchers, similar to the networks of potential reviewers journals have for reviewing manuscripts.

5.2. Trends in the generations and main takeaways

Recent studies indicate that there is not a single LLM that outperforms all others across every task and evaluation scenario. Performance fluctuates considerably according to benchmarks, evaluation approaches, and the LLM-as-a-Judge selected. Various LLMs have distinct strengths and excel in specific tasks. Their rankings can vary according to the evaluation metrics and datasets applied. For example, using the LLM-as-a-Judge approach can introduce biases and might not align with human annotations (Liu, Shi, et al., 2023; Mizrahi et al., 2024).

These are the same conclusions reached in this study. Although human annotators prefer Gemini, the annotator agreement is very low. Overall, the statistical analysis of the *FermBench* dataset shows that, although all evaluated models sustain a strong baseline in factual accuracy and coherence, marked disparities appear when tasks require more advanced reasoning. Gemini and DeepSeek consistently form a distinct *technical tier*, earning statistically higher scores in insightfulness ($p < 0.001$) and comprehensiveness from both human and LLM judges. This indicates that, for complex, fermentation-focused questions, these models deliver a level of integrative, technical synthesis that surpasses the more generalist ChatGPT baseline.

Our two-judge evaluation framework also exposes a key trade-off in LLM assessment. GPT-4o operates as a "style judge", uniquely detecting a highly significant difference in concision ($p < 0.001$), but its pronounced *leniency bias* leads to a ceiling effect that masks pairwise distinctions in overall quality. In contrast, Gemma acts as a "high-resolution judge", offering the necessary evaluative precision to statistically confirm the superiority of Gemini and DeepSeek in overall quality ($p < 0.001$). Collectively, these results imply that, in specialized scientific domains, performance is no longer primarily determined by basic correctness but by the capacity to deliver thorough, insightful and technically rich responses.

In order to better understand the results obtained in this study, we summarize some of the possible advantages and disadvantages of the selected LLMs in Table 9. These advantages and disadvantages are reported by the developers of the LLMs and published in technical reports (Achiam et al., 2023; Anthropic, 2024; Liu et al., 2024; Ramachandran, 2024; Wilkins & Rodriguez, 2024). These specifications can provide more inferential reasoning on why certain models (such as Gemini and DeepSeek) outperform others (e.g., ChatGPT) in this task. For example, ChatGPT is generally statistically and inferentially weaker than all other

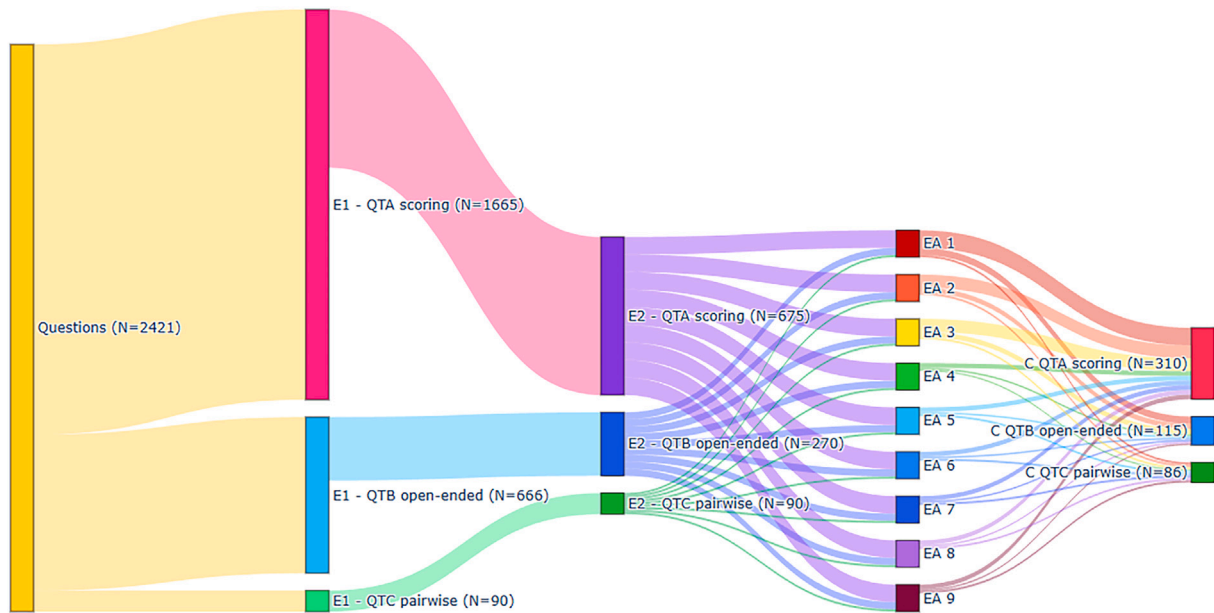


Fig. 5. Sankey plot of the annotation process. For this experiment, we recruited 9 expert annotators (EA). For the first iteration of the experiment (E1), we randomly picked a set of 37 questions, divided into question type A (QTA), which aims to collect scores for the given parameters, question type B (QTB), to collect open-ended questions and question type C (QTC), for the pairwise analysis. The plot, from left to right, shows the total of questions ($N=37 \times 7$), divided into three categories. However, annotators found the task to be too long; therefore, we shortened it to 15 randomly selected questions (E2). In this scenario, if all annotators provided an answer, we would collect a total of QTA=675, QTB=270 and QTC=90. Unfortunately, not all annotators managed to provide answers for all questions. Therefore, the final number of collected answers is, respectively, QTA=310, QTB=115 and QTC=86.

models on complex reasoning, coding, and math problems, while Gemini is more versatile general-purpose model, DeepSeek is built as a “cognitive engine” and exhibits excellent results on reasoning benchmarks (MMLU-Pro) (Wang et al., 2024).

5.3. Educational implications for students, instructors and curriculum design

In spite of the fact that some of the results are not statistically significant, and it was not possible to declare an overall winner among the chatbots, this study is nonetheless relevant to students, as it provides a quantitative analysis of the high quality of the chatbots currently available. This is suggested by the fact that the average score of all models, evaluated by both experts and other LLMs, is substantially higher than average. The results obtained in this study suggest that, in general, students can rely on the responses generated by LLMs in the field of fermentation. However, critical thinking must be used to review the answers, since LLMs might provide incorrect information, as shown in Table 8. Regarding the choice of which chatbot they decide to use, many students do not test all the possible available chatbots, but use the “most famous” model (ChatGPT). However, this study highlights that, often, other tools, such as Gemini or DeepSeek, might be better for insightfulness and overall quality. Moreover, we recommend students to follow institutional and governmental guidelines. In addition to this, given the close scores obtained by the various LLMs used, our study shows that students can also follow personal preferences, perhaps informed by some of the parameters used in the experiment, such as insightfulness and comprehensiveness, “writing” style (e.g., conciseness and readability). However, these results, given the low inter-annotator agreement, should be regarded as preliminary observations and not as a validated pedagogical guidelines.

Furthermore, this study aims to contribute with a robust experimental framework and guidelines that other researchers can follow to perform similar studies. This work presents various metrics and approaches that have been used for evaluating generated text. The experimental framework is shown in Fig. 6. The list below aims to summarize

the experimental framework used, providing strengths and weaknesses for each approach.

- **Automatic metrics**, including cosine similarity, F1 score and BLEU score. These metrics are useful for understanding the responses, their similarity and semantic relationships. However, without a reference text, they are limited in their usefulness.
- **Human annotation (expert scoring)** according to selected parameters on the Likert scale (1-5). Parameters: factual accuracy, comprehensiveness, coherence to the question asked, concision, readability, helpfulness, insightfulness and overall quality. This is a good method to acquire in-depth knowledge of the generated responses and some of their factors; however, it is the most time-consuming approach, and the quality of the annotation can vary.
- **Pairwise comparison** of generated responses. This is a more rapid method for the annotators, since the task is to select the best answer out of two, without having to evaluate every aspect of an answer. On the other hand, some nuances of the responses and their quality are lost with this approach.
- **LLM-as-a-Judge approach** was used due to its rapid spread for evaluating generated text. Among the advantages is the fact that it is relatively inexpensive and fast to perform; however, it has been observed that this method can introduce bias and does not always align with human evaluations.
- **Comparison of generated and curated responses** helps determine whether investing time and resources into data curation is worthwhile, or if generated answers are “good enough”. A disadvantage of this approach is that annotators might not be fully committed to spending the time required to provide a curated answer, making this comparison biased.

We believe that establishing this framework provides a robust methodology for researchers and instructors to evaluate LLM performance within specialized scientific domains. Beyond simple AI tool selection, our findings enable educators to foster GenAI literacy by

Table 9
Summary of the advantages and disadvantages of the selected LLMs.

Model	Primary advantages	Key disadvantages
ChatGPT Resources: Achiam et al. (2023)	Benchmark Baseline & Speed: Serves as the high-quality baseline; still very fast and cheap for simple, high-volume tasks. Highest Accessibility: Widely accessible, and has the most mature API and integration ecosystem in the market. Good Contextual Ability: Highly effective for basic chat, summarization, and quick code generation.	Inferior Reasoning: Statistically and inferentially weaker than all other models on complex reasoning, coding, and math problems. Unimodal Limitation: Primarily a text-only model. Repetitive/Generic Outputs: More prone to repetitive phrasing and less nuanced creative/stylistic outputs compared to the current generation.
Claude Resources: Anthropic (2024)	Elite reasoning & coherence: Exceptional at complex, multi-step reasoning and long-horizon tasks. Agentic and complex workflows: Outstanding at workflows (e.g., RAG, software engineering). High safety and alignment: Strong focus on helpfulness, honesty, and harmlessness; reinforced by research on Constitutional AI, resulting in fewer harmful or biased outputs.	Higher cost: Generally more expensive than models like DeepSeek-V3 or Gemini 2.0 Flash. Slower inference: On some tasks it may be slower than other models. Less multimodal: Compared with some newer or more experimental models, its strengths are more in text.
Gemini Resources: Ramachandran (2024)	Natively multimodal: Can handle images, audio, video, and text natively. Versatile general-purpose model: Good general knowledge, quick prototypes. Speed & throughput: Very fast inference and low latency. Massive Context Window: Supports an extremely large context (up to 1 million tokens), enabling analysis of long documents or codebases.	Shallower Reasoning: While fast, its reasoning depth is shallower than top models (e.g., Claude Sonnet or flagship DeepSeek variants). Weaker on complex or high-stakes tasks: On difficult summarisation datasets, technical domains (e.g., legal, scientific) Ecosystem integration trade-offs: Tight integration with Google services can be a strength or a barrier for non-Google users.
DeepSeek Resources: Liu et al. (2024)	Open architecture and accessibility: Being open-source, DeepSeek-V3 provides transparency, modifiability, and freedom for users. Extreme cost-efficiency: Delivers top-tier performance while substantially reducing costs, highly efficient architecture. Superior math & coding: Statistically robust in mathematical and programming skills; built as a “cognitive engine”. Exhibits excellent results on reasoning benchmarks (MMLU-Pro).	Newer and less established ecosystem: Fewer tools availability, lower API stability and integration maturity. Multimodality limitation: Primarily focused on text and code; multimodal and vision capabilities are not the flagship feature.
le Chat Resources: MistralAI (2025a), Wilkins and Rodriguez (2024)	Highest General-Purpose Performance: Consistently ranks among the top 1–3 models on broad general knowledge benchmarks Strong European Base: Appeals to organizations requiring European-based models. Competitive Pricing: Offers a good balance between high quality and reasonable cost.	Cost vs. DeepSeek: Significantly more expensive than DeepSeek-V3 for similar or slightly lower performance on specialized logical tasks. Multimodality: Primarily a text model; vision capabilities were added later and may not be as integrated as Gemini’s native multimodality.

providing evidence-based guidance on model strengths, e.g., Gemini’s technical insight versus ChatGPT’s concision. By highlighting the distinct *technical tier* of certain models, this work encourages educators to transition from a ‘one-size-fits-all’ AI policy toward domain-specific integration. Ultimately, understanding these performance disparities allows for more intentional curriculum optimization, where AI is used not just as a generalist assistant, but as a specialized partner in complex technical synthesis and problem-solving.

Specific recommendations for educators include integrating LLMs into the curriculum to provide students with feedback before assignment submission; this feedback can focus, for example, on technical richness rather than just grammar. Finally, instructors can design assignments that explicitly require students to compare outputs from different models, teaching critical evaluation of AI reasoning.

5.4. Contributions and overall reflections

Overall, this work contributes to advancing educational research in various aspects. It introduces *FermBench*, a dataset containing 118 questions on fermentation, including curated responses by experts in the field and five sets of responses generated by LLMs powering commercially available chatbots, including ChatGPT, Gemini, DeepSeek, Claude and le Chat. It provides **guidelines to students** regarding which chatbots to use, concluding that the knowledge currently embedded in these models is satisfactory and therefore the decision can be made based on institutional/governmental guidelines and personal preferences. This finding is supported by quantifiable data and reproducible results, as all code and data are made available. It also provides an **experimental framework** that includes various evaluation methods, such as automated metrics, human annotations, and the LLM-as-a-Judge approach. This framework also describes the methods used in detail and discusses the advantages and disadvantages of each, enabling other researchers to replicate this experiment with new data or apply it to other fields.

The study also presents a reflection on the advantages and disadvantages of collecting human annotations. We believe that, by opening the discussion on this topic, the community will be able to find better solutions to alleviate this issue and to ensure curated, and therefore higher quality, data. To do this, we performed an experiment, which is summarized in Table 7. With this experiment, we wanted to compare curated responses and generated ones, to answer the question: *Are custom models actually better?* We found that, in the majority of cases, generated responses were actually preferred. This does not imply that data curation is not necessary; we do believe that this process is fundamental, especially in an educational context. However, we found that often the curated responses were substantially shorter than the generated ones. This might be due to the extensive nature of the task, which led the annotators to focus on the scoring and not the data curation task. A solution could be to divide this process in the future, not to overwhelm the annotators. So, first generate responses, then select the best response, and then ask experts to curate the response and ensure that the information provided is correct.

This is the process selected to improve *FermentAI*: we select the best response for the given question, either among the generated or curated ones, and we ask the expert who annotated the results in Table 7, to curate the selected answers. The data is then used in our custom model, as explained in Section 3.10.

6. Limitations

This study aims to provide an objective comparison of five commercially available GenAI chatbots. It does not account for National and Institutional guidelines regarding these chatbots. If your Country or Institution forbids one or more of these tools, please use the results presented in this study with caution.

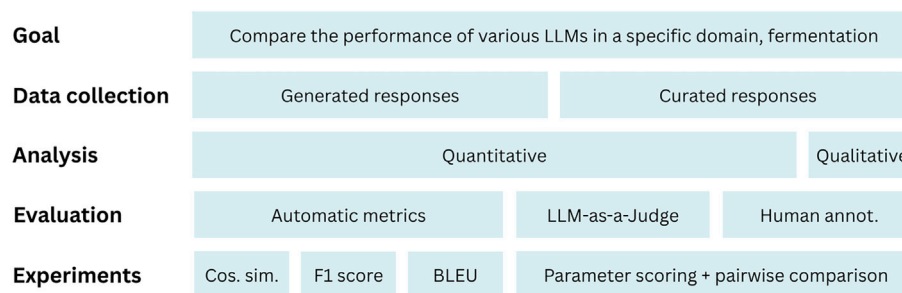


Fig. 6. Summary of the experimental framework used in this study. The figure presents the goal of the study, the type of data collected (both generated and curated by experts), the analysis performed (quantitative and qualitative), the type of evaluation (automatic metrics, LLM-as-a-Judge approach, human annotations) and the experiments performed (including parameter scoring and pairwise comparison).

The choice of selecting GPT-3.5 instead of GPT-4 might seem like a limitation of the study, as the latter model vastly outperforms its predecessor in many tasks. However, the aims of this study are to (a) provide a benchmark of the available free virtual assistants on the topic of fermentation, (b) support students in their learning, and (c) help them choose the most suitable virtual assistant (at least at the time when the code was run, which is April 2025).

It is well known that GenAI chatbots have a sensitivity to prompting. This means that how the prompt is constructed and the additional context provided could greatly affect the generated output. In order to keep this study objective and not favor any particular model, no prompt engineering was performed. We left the default parameters set in the models, and the prompt was composed only of the questions collected in *FermBench*. Although there is some variance in the responses provided by the models, as presented in Table 3, this supports a fairer comparison and ensures reproducibility.

Performing a study that includes human evaluation has some drawbacks. There are some known disadvantages of human evaluation, including its inherent subjectivity, which can lead to inconsistent results, and the high cost of human evaluation. In this study, this is evident from the low inter-annotator agreement ($\alpha = 0.083$), which might raise concerns about the reliability of subjective scoring and generalizability of the conclusions. The results presented in this study should be interpreted as preliminary observations and not as validated pedagogical guidelines. A potential mitigation for the low inter-annotator agreement and, therefore, leading to more reliable results would be to recruit a larger annotator pool in the future or perform multi-round calibration. However this would require creating an incentive to accelerate the process. Offering compensation may spur annotators to devote maximum focus and meticulousness, thus helping to ensure consistent, high-quality data. In this work, annotators were volunteers recruited from the professional network of the authors.

GenAI chatbots (and the LLMs behind them) are constantly evolving and undergoing improvements. Educational research often struggles to keep up with the pace of these changes, making it difficult to conduct timely and relevant studies and capture the nuances of new learning environments (Lavicza et al., 2022). However, in order to perform robust educational research, time is needed. Within the six months necessary to start the project, collect the data, perform the experiment, analyze the data and write the manuscript, all the chatbots used in this study had changed the underlying LLM. Therefore, although the numerical results reported in this study could potentially be outdated by the time the article is published, the purpose of this work will not be obsolete. In fact, this work aims to: (i) initiate a discussion around the selection of different GenAI chatbots; and, (ii) provide an evaluation framework that can be applied to benchmark different models. Additionally, to ensure the manuscript remains useful in the future, the benchmark dataset (*FermBench*) and code are open-source on GitHub (<https://github.com/FiammettaC/FermBench>). The repository will continue to be maintained and enriched, and users can submit a request for results to be included.

7. Conclusion

This work aims to contribute to the advancement of educational research by answering two questions: whether the current knowledge of LLMs is good enough for higher education and, in case it is, if students and educators can freely choose the model of their preference. This study introduced *FermBench*, a curated dataset that includes 118 fermentation-related questions and both curated and generated responses. These responses were generated using five LLMs employed in commercially available chatbots: ChatGPT, Gemini, Claude, DeepSeek and le Chat. To evaluate the generated answers, we used various approaches. This culminated in a robust experimental framework that includes automated metrics, human annotations, and the LLM-as-a-Judge method. To provide directions to other researchers aiming to replicate this work on different domains, we thoroughly described the methods used and discussed the strengths and weaknesses of each approach.

The analysis reveals that while most models handle basic facts well, Gemini and DeepSeek distinguish themselves in a “technical tier” by providing significantly deeper insight and comprehensiveness than ChatGPT. Furthermore, the study highlights that evaluation results depend heavily on the judge: GPT-4o acts as a “style judge” that overemphasizes concision and leniency, whereas Gemma serves as a “high-resolution judge” capable of detecting the superior technical quality of the top-performing models. Ultimately, the findings suggest that excellence in specialized scientific fields is now defined by technical richness rather than just basic factual correctness.

However, given the overall high baseline and the limited agreement between the annotators, as well as the fact that the judges were not able to agree on one overall best model, we concluded that the current knowledge embedded within these models is adequate and standalone results cannot provide pedagogical guidelines regarding which chatbot should be used in education. These results suggest that the choice of which GenAI chatbot should be supported by institutional or government guidance, as well as individual preferences, perhaps supported by the parameters identified in our analysis.

A limitation of this type of work is that LLMs continuously improve, and often educational research is not able to keep up. To tackle this issue, we publish all the code and data, with the resolution to frequently update the leader-board of models benchmarked on *FermBench* and, therefore, provide an updated point of reference for students wishing to learn fermentation.

8. Future perspectives

We will continue to improve *FermBench* (Caccavale et al., 2024). To contribute to the model’s improvement, it is possible to fill out the form available at <https://forms.gle/ikAyxfE5EBwcu51S8> with your suggested questions.

Due to the instability of single-prompt evaluations and the need for multi-prompt or thorough benchmarking (Mizrahi et al., 2024),

future experiments will be performed using *FermBench* in a multi-prompt evaluation approach.

CRedit authorship contribution statement

Fiammetta Caccavale: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization
Adem R.N. Aouichaoui: Writing – review & editing, Validation, Software, Investigation, Formal analysis, Data curation
Ulrich Krühne: Writing – review & editing, Supervision, Resources
Krist V. Gernaey: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition
Carina L. Gargalo: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition

Funding

Funding for this study was provided by (i) the European Union under [Horizon Europe](#) research and innovation programme with the grant agreement [101159993](#), within the HORIZON-WIDERA-2023-ACCESS-02-0 call (Dig4Bio); (ii) [Novo Nordisk Foundation](#), the grant number is: [PNNF22OC0080136](#) - Real-time sustainability analysis for Industry 4.0 (Sustain4.0); and (iii) Technical University of Denmark (DTU).

Ethics statements

Informed consent was obtained from all participants and their privacy rights were strictly observed. Before participating, all participants were informed that the collected data would be used for research purposes and would be handled anonymously. Detailed information on the study was provided to the participants before starting the survey, reassuring them of their right to voluntary participation and withdrawal. The data can be obtained by sending an email request to the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank the experts for the essential annotation and curation work performed in this study. We greatly appreciate the effort put into performing the task and the time dedicated to it.

Appendix A

The appendix contains supplementary information which may be helpful in providing a more comprehensive understanding of the work performed. [Figs. 7 and 8](#) show the semantic relationships across the five groups in a shared space. As in [Fig. 3](#), we create a single array that contains all the sentence embeddings from all five response datasets. Then we applied the dimensionality reduction technique to the combined dataset. This approach focuses on finding the dimensions that best capture the overall variance and structure across the datasets, therefore resulting in a single transformed space where all sentences are embedded.

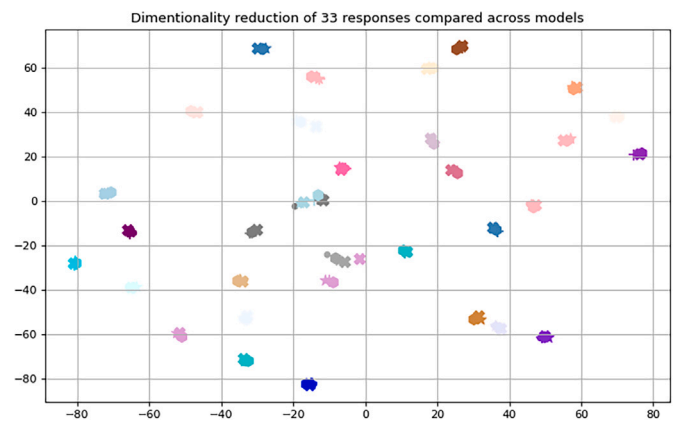


Fig. 7. Dimensionality reduction using the T-SNE method for the responses of the selected models. ChatGPT is represented by the “X” symbol, Claude by the hexagon, Gemini by the dot, DeepSeek by the plus sign and le Chat by the star.

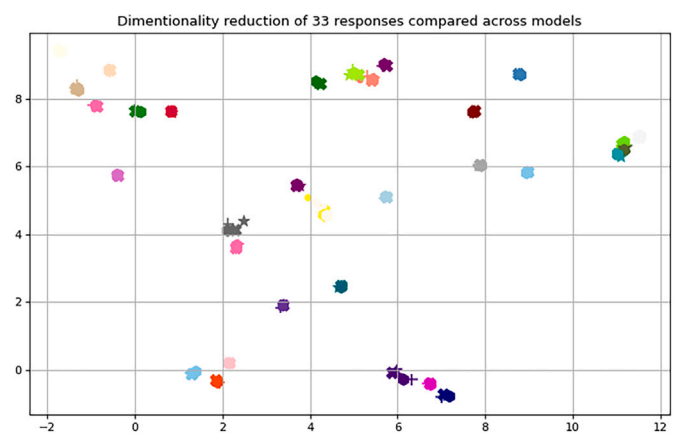


Fig. 8. Dimensionality reduction using the UMAP method for the responses of the selected models. ChatGPT is represented by the “X” symbol, Claude by the hexagon, Gemini by the dot, DeepSeek by the plus sign and le Chat by the star.

Appendix B. Supplementary data

Supplementary data for this article can be found online at doi:[10.1016/j.caeai.2026.100577](#).

Data availability

This study is committed to the open-source set of principles. The code, together with *FermBench*, the curated fermentation dataset, is freely available on GitHub at <https://github.com/FiammettaC/FermBench>. The code is published under the MIT license. Currently, it is compatible with Python 3.x, for Windows and Linux operating systems. It is also recommended to use a conda environment to run the code. Other data can be obtained by sending an email request to the corresponding author.

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